Evidence Board · Phase 2, EXPLORE

Which Yucatán municipalities show groundwater quality flags, and how much of the state does the monitoring network see?

In short

12 of the 123 groundwater sites (9.8%) are flagged by CONAGUA as unsuitable as a drinking-water source, in 10 of the 65 monitored municipalities. **28 sites (22.8%)** are restricted for irrigation, in 16 municipalities. And **41 of Yucatán's 106 municipalities have no groundwater site in the CONAGUA files**, so for them the data says nothing, good or bad.

9.8% drinking-source flag
12 of 123 groundwater sites, 10 municipalities

22.8% irrigation flag
28 of 123 groundwater sites, 16 municipalities

41 municipalities, no site
of 106 in Yucatán have no groundwater site in the files

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1. Data used

Source	CONAGUA, RENAMECA: 30 Excel files, one per parameter, sheet "2012-2022" in each.
Size	2,106 rows, 147 monitoring sites (CLAVE), 70 municipalities.
Analysed	Groundwater: 14 files, 123 sites in 65 municipalities. Five parameters give the flags: nitrates, arsenic, fecal coliforms, conductivity and dissolved solids (SDT).
Not analysed	Surface water: 16 files, 24 sites in 11 municipalities.
Period	2012–2022 (the sheet name). One value per site and parameter for the whole period.
Denominator	106 municipalities in Yucatán (INEGI).

2. Preparation and data quality

Check	What we found	What we did
Merge	30 files with the same 9 columns, one sheet each, 2,106 rows.	Stacked them, keeping the file name and sheet name. The row count of every file was re-counted independently and matched (2,106 = 2,106).
Values stored as text	Numbers are text in the source. 883 rows read "<limit" (below detection); 195 rows have no value; every value present could be read.	Parsed to numbers; kept the original text and flagged below-detection rows. Nothing was imputed.
Rows with no value	All 195 rows with no value are in surface-water files. Groundwater files are complete (123 of 123 sites).	No action needed for the groundwater analysis.
Names and duplicates	No duplicate site-and-parameter pairs. No site with two names or two municipalities; the 70 municipality names are clean.	None needed.
Shared parameter code	Both coliform files use the code `coli_fec`.	Used the file name as the parameter key.
Mislabelled file	The file named "Magnesio" has code `mn` (manganese). CONAGUA's list of groundwater indicators has manganese (Manganeso total) and no magnesium.	Left it out; it is not part of the question.
Below detection	Arsenic 115 of 123 values and fecal coliforms 85 of 123. Cadmium 123, lead 123 and mercury 122 of 123.	Flags use CONAGUA's label. Cadmium, lead and mercury have no variation and are left out.
Missing information	No sampling dates, no coordinates, no units in the files.	Analysis kept at site and municipality level; no trends; units not reported.

3. Procedures

1. **Water type.** A site is groundwater if it appears in the groundwater coliform file and surface water if it appears in the surface coliform file. The 14 groundwater files share the same 123 sites and the 16 surface files the same 24, so the two groups do not overlap.
2. **Flag.** A site is flagged when CONAGUA's own label in the Calidad column is one of those in the table below. We added no threshold. CONAGUA judges groundwater by suitability as a drinking-water source and for irrigation (CONAGUA, Indicadores de Calidad del Agua), which is why we use these two uses. For irrigation we read the labels as restrictions because the values rise with the label: for SDT, "sensibles" reaches 986, "manejo especial" starts at 1,010 and "tolerantes" at 2,082.
3. **Counts.** Flagged sites per parameter, per use (drinking = any of three parameters, irrigation = any of two) and per municipality.
4. **Coverage.** Sites per municipality, and the 106 − 65 = 41 municipalities with none.

5. **Sensitivity.** Alkalinity also has a drinking-source label ("Indeseable como FAAP"). Counting it would add 2 sites (14 instead of 12). We kept to the three parameters in the question.

Parameter	Use	CONAGUA label counted as a flag	Flagged / sites	Value range of flagged
Nitrates	Drinking	No apta como FAAP	3 / 123	11.5–13.9
Arsenic	Drinking	No apta como FAAP	2 / 123	0.032–0.049
Fecal coliforms	Drinking	Contaminada	7 / 123	1,100–3,076
Conductivity	Irrigation	Dudosa para riego or Indeseable para riego	20 / 123	2,038–6,093
Dissolved solids (SDT)	Irrigation	Cultivos con manejo especial or Cultivos tolerantes	28 / 123	1,010–4,504

4. Results

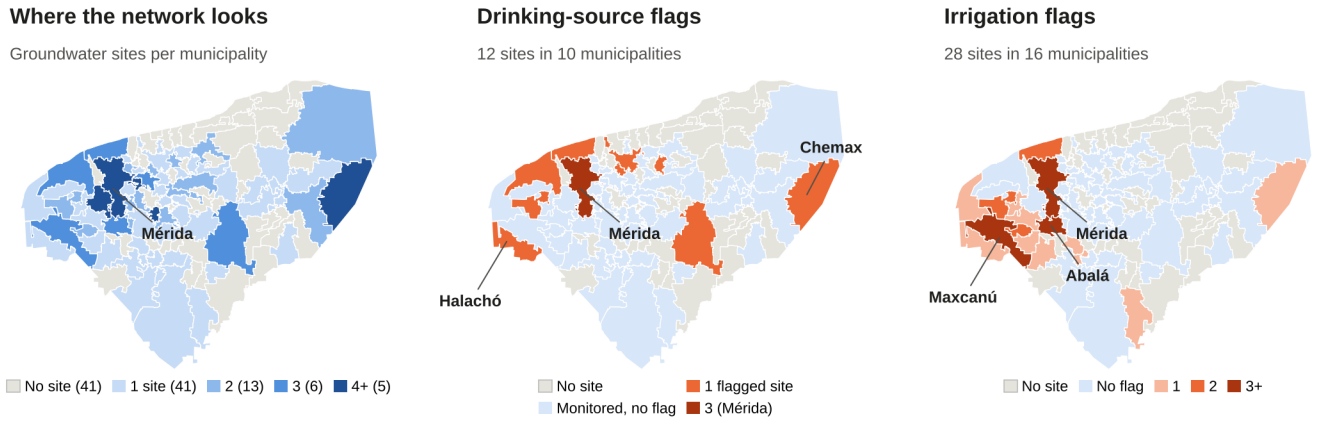


Figure 1. Groundwater sites, drinking flags and irrigation flags by municipality. Colour is by municipality, not by site, because the files have no coordinates. Boundaries: INEGI Marco Geoestadístico 2025. The offshore islets of Progreso and Celestún are outside the view.

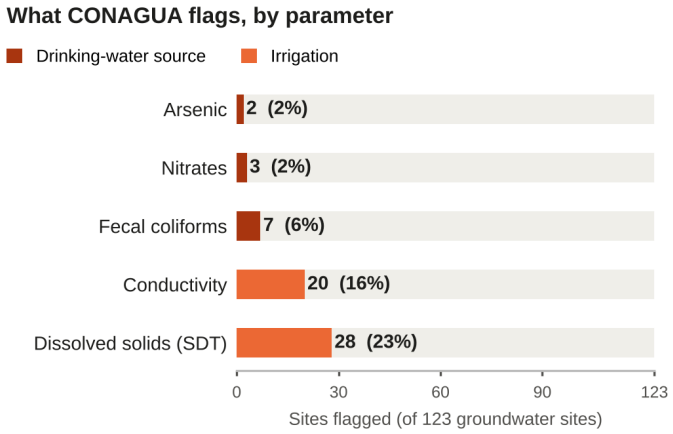


Figure 2. Groundwater sites flagged, by parameter. Share is of 123 sites. 12 sites carry a drinking flag and 28 an irrigation flag.

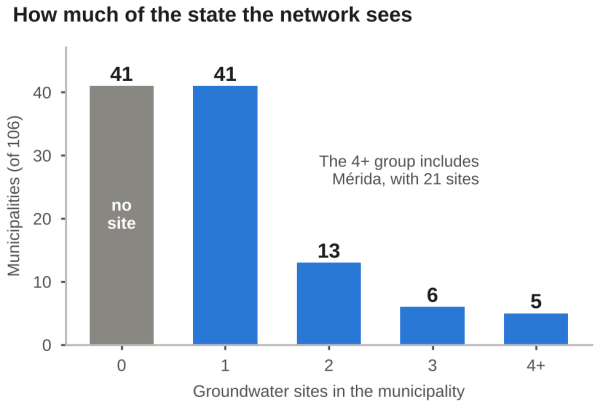


Figure 3. Municipalities by number of groundwater sites. The grey bar is the 41 municipalities (of 106) with no site in the files.

Overlap of the two kinds of flag. 5 sites carry both, 7 only a drinking flag, 23 only an irrigation flag and 88 none. In all, 35 sites (28.5%) in 21 municipalities carry at least one flag.

Table 1. The 10 municipalities with at least one drinking-flag site. Counts of sites, not rates.

Municipality	Sites monitored	Drinking-flag sites	Irrigation-flag sites	Parameters flagged
Mérida	21	3	6	Arsenic, Coliforms, Conductivity, SDT
Chemax	4	1	1	Arsenic, SDT
Hunucmá	3	1	0	Coliforms
Progreso	3	1	2	Coliforms, Conductivity, SDT
Yaxcabá	3	1	0	Coliforms
Kinchil	2	1	2	Coliforms, Conductivity, SDT
Muxupip	2	1	0	Coliforms
Halachó	1	1	1	Nitrates, Conductivity, SDT
Motul	1	1	0	Nitrates
Tepakán	1	1	0	Nitrates

5. Finding

Observed (what the data shows)

- 12 of 123 groundwater sites (9.8%) carry a drinking-source flag: 7 for fecal coliforms, 3 for nitrates and 2 for arsenic. No site is flagged for two of these parameters. They are in 10 municipalities: Mérida has 3 and each of the other nine has 1.
- 28 sites (22.8%) carry an irrigation flag, in 16 municipalities. All 20 sites flagged for conductivity are also flagged for SDT.
- 65 of 106 municipalities have a groundwater site in the files. 41 (38.7%) have none, and 41 of the 65 have exactly one. Mérida holds 21 sites (17.1%).

Interpretation (our reading)

- Irrigation restrictions are more common than drinking-source flags. They come from two salt-related parameters (SDT and conductivity), while the drinking flags come from three different parameters with few sites each.
- Because most municipalities have one site, a flag describes one point, not the municipality. Where there is no site there is no evidence either way, so "no flag" must not be read as "good water".
- Mérida shows more flags in absolute terms partly because it has far more sites.

Implication (what it could mean for the audience)

- Any statement at municipality level needs more than one site per municipality. The coverage gap is a finding on its own, not only a limit.
- The 12 drinking-flag sites are a short list where follow-up sampling is easiest to justify. The dashboard will let a reader filter to them.

6. Limits

- No sampling dates: no trends or seasons. The values are probably summaries of 2012–2022, but the files do not say so, so we do not call them averages.
- CONAGUA's labels are used as published. A flag is not a health or crop diagnosis. Our reading of the irrigation labels as restrictions comes from the label text and the order of the values.
- Most values for arsenic (115 of 123) and fecal coliforms (85 of 123) are below detection, so flags exist only where a detectable value was reported. Units are not in the files.
- The network is not a random sample, and one site stands for most municipalities. We report counts, not rates. The maps colour whole municipalities, since the files have no coordinates.
- Alkalinity would add 2 drinking-flag sites if counted. Surface water, and the cadmium, lead, mercury and "Magnesio" files, are outside this analysis.
- The overlap of drinking and irrigation flags (5 sites) is descriptive. We make no causal claims.

7. Files to verify the analysis

Everything above can be reproduced from the files in this folder: **merge_xlsx.py** (merges the 30 files and recounts rows), **profile_agua.py** (quality tables and the tidy table), **evidence_analysis.py** (flags, coverage, figures), **merged_calidad_agua.xlsx**, **datos_tidy.csv** and **evidence_tables.xlsx** (every table behind this report). Commands:

```
python merge_xlsx.py DATA_FOLDER merged_calidad_agua.xlsx
python profile_agua.py merged_calidad_agua.xlsx perfil_out
python evidence_analysis.py perfil_out/datos_tidy.csv evidence_out
```

AI use

AI use. Claude (Anthropic) was used to write the merge, profiling and analysis scripts, to run the data checks and to help draft this document. The scripts are in the team folder so every number can be reproduced.

References

1. CONAGUA, RENAMECA (Red Nacional de Medición de la Calidad del Agua). 30 state-level Excel files, one per parameter, named `Calidad_de_agua_Estado_<parameter>.xlsx`, sheet "2012-2022".
2. Instituto Nacional de Estadística y Geografía (INEGI) (2021), Panorama sociodemográfico de Yucatán, Censo de Población y Vivienda 2020 (list of the state's municipalities, keys 001 to 106): https://en.www.inegi.org.mx/contenidos/productos/prod_serv/contenidos/espanol/bvinegi/productos/nueva_estruc/702825198046.pdf
3. CONAGUA, Red Nacional de Medición de la Calidad del Agua (RENAMECA), official presentation (groundwater indicators, including manganese, and the quality labels such as "Dudosa para riego" and "Cultivos tolerantes"): https://files.conagua.gob.mx/transparencia/CalidaddelAgua.pdf
4. CONAGUA, Indicadores de Calidad del Agua (groundwater sites are evaluated on suitability as a drinking-water source and for agricultural irrigation), consulted 13 November 2026: https://www.gob.mx/conagua/articulos/indicadores-de-calidad-del-agua
5. Ríos-Ponce, J. D., Acosta-González, G. and Cejudo, E. (2020), La precipitación histórica y la extracción del agua subterránea en la península de Yucatán: una reflexión, Desde el Herbario CICY 12: 110–118: https://cicy.repositorioinstitucional.mx/jspui/bitstream/1003/2588/1/2020-06-04-Rios-Ponce-et-al-La-precipitacion.pdf
6. Instituto Nacional de Estadística y Geografía (INEGI), Marco Geoestadístico 2025, municipal boundaries of Yucatán (state 31), file `2025_1_31_MUN`; used only to draw the maps.